

Jinxi (Jace) Yu

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EDUCATION

University of California, Los Angeles (UCLA)

B.S. in Computer Science

GPA: 3.97/4.00 | Awards: Dean's Honors List

Relevant Coursework: Object-Oriented Programming (C++), ML, NLP, DL, Algorithms, Operating System, Data Mining

Los Angeles, CA

Sep 2024 – Jun 2027

RESEARCH & WORK EXPERIENCE

UCLA Moonshot Math Harness (Prof. Kai-Wei Chang, Nanyun Peng, Terence Tao)

Research Assistant

Los Angeles, CA

Jul 2026 – Present

- Built a **long-horizon multi-agent reasoning harness** for autonomous mathematical research that has **solved multiple open mathematical problems**, orchestrating specialized explorer, verifier, and consultant agents over a rich library of **skills and MCP tools**, with per-role capabilities **statically enforced** by the harness rather than governed by prompts
- Designed a **persistent graph-based memory system** maintaining an evolving **exploration graph** across long-running executions — preserving intermediate results and competing proof directions — with **self-evolving** memory curation and **iterative planning** that prioritize high-value branches and adapt problem-solving strategy

NLP Lab UCLA (Natural Language Processing)

Research Assistant

Los Angeles, CA

Mar 2026 – Present

- Built a privacy-preserving safety guard for LLM-based multi-agent systems, training a **graph attention network** over inter-agent communication graphs via **federated learning** to detect and block risky agents without sharing raw interaction traces
- Matched centralized-training detection quality (within **0.03 AUROC**) across **5** agent-safety benchmarks, and cut ground-truth attack-success rate by **43%** in live deployment as a lightweight CPU sidecar; first-author paper in submission to **AAAI 2027**
- Accelerated LLM MAS topology generation by replacing graph diffusion generator (GTD, ACL 2026) with **VQ-VAE**, and distilled the **GNN** reward proxy into an **MLP** via graph vector quantization, achieving **64×** speedup at task performance on par with the baseline

SCAI Lab UCLA (AI and Data Mining)

Research Assistant

Los Angeles, CA

Feb 2026 – Present

- Developed **Gaussian speculative decoding** in **PyTorch**, pairing lightweight draft models with **single-pass parallel verification** to accelerate inference of pretrained time series foundation models by up to **3.1×** across **5** model families without retraining
- Proved a **capped-geometric throughput law** making speedups predictable before deployment, and a **bounded-deviation guarantee** on forecast fidelity to the target model; first-author paper in submission to **AAAI 2027** (arXiv:2511.18191)

Yiyue Intelligent Technology Co., Ltd.

Full-Stack Developer Intern

Nanjing, China

Jul 2025 – Sep 2025

- Developed key modules (grade & course management, role-based notifications) for a university web using **Vue.js** and **Java/Spring Boot**
- Used **Spring Data JPA** to perform CRUD operations on **MySQL** without raw SQL, ensuring data integrity across **50+** test cases

Huatai Securities Co., Ltd. (Wealth Management Dept.)

Data Analysis Intern

Nanjing, China

Jun 2025 – Jul 2025

- Analyzed **2,000+** high-net-worth client records to identify potential investment leads using structured financial metrics
- Cross-referenced foundation and university financial statements to identify **30+** organizations with investable endowment assets

PROJECT HIGHLIGHTS

EgoX - AI-Powered Personal Career Wiki | Next.js, Supabase (Postgres + pgvector), Vercel AI SDK, Drizzle ORM

Sole Developer

Apr 2026 – Present

- Built a **multi-tenant AI career-wiki SaaS** that converts raw career material into a structured knowledge base via **LLM structured extraction** (Zod-constrained outputs), powering **grounded RAG chat** with inline citations (**HNSW** vector search + knowledge-graph expansion), source-traceable resume generation, and voice mock interviews (**TTS/STT**) with **LLM-as-judge** rubric scoring
- Engineered a **sealed LLM gateway** metering every call end-to-end: **atomic budget reservations** via single conditional UPDATE (eliminating check-then-act races), full cost audit trail, **tiered model routing** ($\sim 5\times$ cost cut), and **batched embeddings** cutting ingest latency **15s** $\rightarrow$ **600ms**; shipped **Stripe** billing with idempotent webhooks, **Postgres**-backed rate limiting, and server-side **multi-tenant isolation**

Mock-Trader | Trading Simulation Platform & Agent Evaluation Sandbox

Sole Developer

Dec 2025 – May 2026

- Built a trading platform with **Node.js/TypeScript**, **PostgreSQL** trade ledgers, position reconstruction, rule engines, live market data
- Open-sourced an **npm MCP server** exposing **16 tools** for AI trading-agent evaluation: order execution with fees/slippage, **prop-firm-style risk rules**, and **strict replay backtests** on Alpaca/Polygon/CSV data, hiding future bars **server-side** to prevent **look-ahead bias**

SKILLS

Languages: C++, Java, Python, Go, JavaScript, TypeScript, HTML/CSS

Frameworks/Tools: PyTorch, CUDA profiling, Node.js, React, Vue, Redis, Pandas, Spring Boot, Bash, Git, MySQL, Docker, PostgreSQL